

# Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

**Input Content definition:** Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

Input content is the highest-priority dimension per elicitation (HIGH), and dataset analysis reveals a critical concern: all 26 sampled Hindi validation examples are flagged is\_translated=True, suggesting the Hindi validation split is dominated by GPT-4O-translated English-source items rather than natively-sourced Hindi competitive-exam questions. While MILU's documentation claims an India-first design [Q11] and includes regionally relevant subjects [Q12, Q14], the actual Hindi content sampled shows: (a) literary content focused on Wordsworth rather than Tulsidas/Premchand/Kabir [DATASET-D5]; (b) translated technical terminology using transliterations (शेल, नाइट्रेट, जंप लेबल) that may not align with state-board Hindi textbook conventions [DATASET-D14, D15, D23]; (c) only one item with explicit North Indian state-specific content (UP minerals) [DATASET-D2]; (d) parallel translated items appearing across all 11 language configs [DATASET-D11, D18, D19, D24]. The paper's dataset-level 25% translation figure [Q47] underestimates the apparent Hindi-validation-split translation rate.

| ID  | Evidence                                                                                                                                                                                                                      |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q11 | "We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)                                                                                            |
| Q12 | "These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)                                                          |
| Q14 | "We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)                                                                                                                |
| Q47 | "Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)                                                                       |
| D2  | नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question                                                                                      |
| D5  | विलियम वर्ड्सवर्थ _____ के कवि हैं। इंग्लैंड — Literature question about English poet Wordsworth — not Hindi-board canonical literature                                                                                       |
| D11 | What was the first capital of the Bahamani Kingdom? ... Gulbarga — Medieval Indian history — source question in English                                                                                                       |
| D14 | स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration                                                                                  |
| D15 | रक्त में नमिनलखिति में से किसकी अधिकता 'ब्लू बेबी सड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained                                                           |
| D18 | What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1      |
| D19 | 1991 में वित्तीय संकट को द्रगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check |
| D23 | फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉरमेट लेबल — Computer science MCQ — translated, technical vocabulary                                                        |
| D24 | एक भौगोलिक क्षेत्र में नमिनलखिति वशिष्ट वशिषताएँ हैं: ... उत्तरी अमेरिका का अटलांटिक तट — Mediterranean climate geography — translated, no North India content                                                                |

## Strengths

- Documented India-first design intent with stated coverage of local history, arts, festivals, and laws [Q2, Q11, Q12]
- Region-specific exams are explicitly emphasized for capturing local knowledge [Q14, Q28]

- Some India-centric Hindi items present (UP minerals, Mughal history, Amrita Shergil, Radcliffe Line) [DATASET-D2, D3, D8, D25]
- Cross-language answer-key consistency demonstrated for translated parallel items [DATASET-D18, D19]

| ID  | Evidence                                                                                                                                                                                                                    |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q2  | “With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science.” (p.1)    |
| Q11 | “We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.” (p.2)                                                                                          |
| Q12 | “These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)                                                        |
| Q14 | “We focus on region-specific exams to authentically capture local knowledge in the respective language.” (p.2)                                                                                                              |
| Q28 | “Regional state exams are particularly valuable as they cover various state-level topics and emphasize the official language of each state.” (p.3)                                                                          |
| D2  | नमिन्लखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question                                                                                   |
| D3  | मुगल ग्रंथ 'मासिरि-ए-आलमगीरी' के रचयिता कौन हैं? साकी मुसतईद खान — Mughal history — competitive-exam GK register                                                                                                            |
| D8  | नमिन्लखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार हैं, जिन्हें कभी-कभी भारत की फ्रीडा काहलो के नाम से जाना जाता है? अमृता शेरगिल — Indian arts & culture — Amrita Shergil, India-centric                                    |
| D18 | What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? Sharp rise in value of imports of oil & petroleum products — 1991 India economic crisis — same question as Bengali D1    |
| D19 | 1991 में वित्तीय संकट को दूर करने वाले विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of English Ex. 1 — translation quality check |
| D25 | रेडक्लिफ रेखा भारत की सीमाओं को नमिन्लखिति में से किस देश के साथ विभाजित करती है? पाकिस्तान — Partition history — India-centric GK                                                                                          |

## Checklist

### Checklist item definitions:

| ID   | Assessment Question                                                        |
|------|----------------------------------------------------------------------------|
| IC-1 | Do inputs require region-specific cultural/geographic/dialectal knowledge? |
| IC-2 | Does culturally sensitive content align with target culture?               |
| IC-3 | Flag inputs assuming Western-specific knowledge                            |
| IC-4 | Regional annotator recruitment for culturally sensitive instances          |
| IC-5 | Document content issues harming content validity                           |

### Checklist responses:

**IC-1:** The deployment requires region-specific knowledge: North Indian state geography, Hindi-board canonical literature, Panchayati Raj civics, state freedom-movement figures. The Hindi sample contains some India-centric items [DATASET-D2, D3, D25] but a majority of items are translated [DATASET-D28] and may not reflect native Hindi competitive-exam content.

**IC-2:** Cultural alignment is partially present (Mughal history, Indian arts, partition geography are reasonable matches [DATASET-D3, D8, D25]), but the Hindi-literature register [DATASET-D5 — Wordsworth as the only literature item] does not match Hindi-medium school-board norms. Canonical Hindi literary figures (Tulsidas, Premchand, Kabir, Mahadevi Varma) are not present in the sampled items.

**IC-3:** Translated items retain English technical transliterations (शेल, नाइट्रेट, प्लूसेटिंग डीसी, जंप लेबल) [DATASET-D14, D15, D23] that may diverge from Vaigyaanik evam Takniki Shabdavali Aayog or state-board conventions. The Wordsworth literature item [DATASET-D5] introduces non-regional content. Translated items effectively test translated-English knowledge rather than native Hindi-register knowledge.

**IC-4:** INSUFFICIENT DOCUMENTATION — MILU does not document recruitment of Hindi-medium North Indian annotators specifically [NOT FOUND per regional YAML annotator\_demographics]; institutional base is IIT Madras + IBM Research India, which does not guarantee North Indian Hindi-medium representation.

**IC-5:** Documented content issues: (a) ~100% translation rate observed in Hindi validation sample vs. ~25% claimed dataset-wide [Q47, DATASET-D28]; (b) absence of canonical Hindi-board literary content; (c) under-representation of UP/Bihar/Rajasthan/MP state-specific civics; (d) translation terminology not validated against state-board Hindi norms [NOT FOUND per regional YAML hindi\_register\_standardisation].

| ID     | Evidence                                                                                                                                                                                                                                              |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q11    | "We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)                                                                                                                    |
| Q12    | "These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)                                                                                  |
| Q14    | "We focus on region-specific exams to authentically capture local knowledge in the respective language." (p.2)                                                                                                                                        |
| Q47    | "Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)                                                                                               |
| Q84    | "Third, the scarcity of questions necessitated translating a portion of the dataset." (p.9)                                                                                                                                                           |
| D2     | नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — North Indian state-specific geography: UP minerals question                                                                                                              |
| D3     | मुगल ग्रंथ 'मासरि-ए-आलमगीरी' के रचयिता कौन हैं? साकी मुस्तईद खान — Mughal history — competitive-exam GK register                                                                                                                                      |
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| D8     | नमिनलखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार थीं, जिन्हें कभी-कभी भारत की फर्दा काहलो के नाम से जाना जाता है? अमृता शेरगलि — Indian arts & culture — Amrita Shergil, India-centric                                                                |
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| D25    | रेडक्लफि रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ वभिजति करती है? पाकस्तान — Partition history — India-centric GK                                                                                                                       |
| D28    | जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... is_translated: True — is_translated=True flag on all 26 Hindi validation examples reviewed                                                                                                              |
| WEB-19 | Bihar's majority population speaks Bhojpuri (24.86%), Maithili (12.55%), Magahi (10.87%) — Khari Boli register may amplify mismatch for this sub-population ( <a href="https://en.wikipedia.org/wiki/Bihar">https://en.wikipedia.org/wiki/Bihar</a> ) |

## Information Gaps

- Whether the test split is also predominantly is\_translated=True (only validation split sampled)
- Hindi-specific translation proportion in MILU (paper reports only dataset-wide 25%)
- State-wise distribution of Hindi exam sources (UP/MP/Rajasthan/Bihar shares)
- Whether Hindi literature items in the un-sampled portion include canonical board authors

## Requires Expert Verification

- Whether GPT-4O Hindi translations preserve subject-specific terminology in line with UP/MP/Rajasthan/Bihar state-board textbooks
- Whether Khari Boli register mismatches with Bhojpuri/Maithili-speaking Bihar teachers materially affect comprehension
- Whether 'शेल', 'नाइट्रेट', 'जंप लेबल' style transliterations are acceptable in state-board Hindi technical texts

## Remediation

**Gap 1: 100% of sampled Hindi validation items are machine-translated; translated technical terminology uses English transliterations of unvalidated alignment with state-board Hindi norms; canonical Hindi-board literary content absent from sample**

**Recommendation:** Stratify any deployment-relevant evaluation by is\_translated flag and report performance separately for natively-sourced vs. translated Hindi items; supplement with a curated set of native Hindi-board MCQs covering Hindi

literature canonical authors and verified state-board terminology

**Gap 2: No published state-wise breakdown of Hindi exam sources for UP/MP/Rajasthan/Bihar**

**Recommendation:** Request access to the gated HuggingFace dataset metadata (or supplementary Table 8) and audit the Hindi item provenance by source-exam state; if Hindi-belt state-PSC sourcing is thin, supplement with state-PSC items directly